

SPACE X

Mission Notebook · Interplanetary Spacecraft & Satellite Engineering

Four things to remember. Everything else in here is detail.

🖊 DRAW IT FIRST

❓ CUES = QUESTIONS

🧠 SUMMARY, VIDEO OFF

🔁 COME BACK +1 +3 +7

🖊 COLOR AFTER, NOT DURING

Personal study notebook. Not affiliated with any company.

STEP 1 · 10 MIN · ONCE

Set up the notebook before Lecture 1

Four reserved spots. Do this once, tonight, then never think about it again.

🔁 INSIDE FRONT COVER · REVIEW TRACKER

Section	Watched	+1	+3	+7	+14	+30
1 Mission design	11/7 ✓	✓				
2 Environment	11/8 ✓					
3 Orbital mech						
4 Propulsion						
5 Comms						
6 ADCS						
7 Power						

A tick = 5 min: cover notes, answer the cues out loud. Ticks are the score.

📖 PAGES 1–2 · INDEX

Index **pg**

Sec 1 / Lec 3 Mission design 5
Sec 1 / Quiz 1 Time of flight 7
Sec 2 / Lec 1 Vacuum & thermal 9
Sec 2 / Lec 2 Radiation 11

One line per lecture. Write it when you START the spread.

🧮 PAGES 3–4 · EQUATIONS & KEY NUMBERS

Equations & key numbers **UNITS!**

$T = 2\pi \sqrt{(a^3/\mu)}$ period · μ_{Earth}=398,600 km³/s²

$TOF = \pi \sqrt{(a^3/\mu)}$ Hohmann = HALF period · LEO→GEO 5.3 h

$v = \sqrt{(\mu/r)}$ circular · 400 km → 7.67 km/s

$v^2 = \mu(2/r - 1/a)$ vis-viva · any point, any orbit

GEO = 35,786 km · LEO ≈ 200–2000 km · Mars window every ~26 months · Mars radio delay 4–24 min

Formulas boxed with ONE worked number. No formula in a lecture? Put its key NUMBERS here instead. Nothing else on these pages.

❓ LAST 10 PAGES (from the back) · QUESTION BANK

Q — this page **A — facing page**

1. Why is TOF half the period?
2. Fastest at perigee or apogee? Which law?
3. Why no fan to cool a satellite?
4. ? "C3" — lost me → ask Claude

Every miss + every "?" lands here. This is your flashcard deck. Answers on the facing page so you can cover them.

STEP 2 · EVERY LECTURE

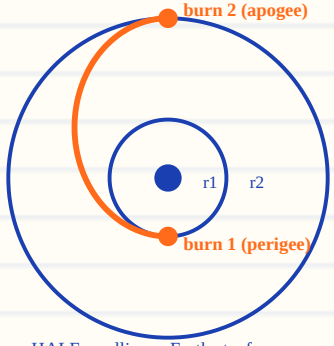
One lecture = one two-page spread (filled example)

Grey = your guess before the instructor shows it. Blue = corrected after. Example: Section 3, Hohmann transfer.

LEFT PAGE · DRAW

Sec 3 / Lec 2 — Hohmann transfer **11/9 · pg 13**





HALF an ellipse · Earth at a focus

Worked: LEO 300 km → GEO 35,786 km
r1 = 6371+300 = 6671 · r2 = 6371+35786 = 42157 (km, from CENTER)
a = (6671+42157)/2 = 24,414 km
a^3/μ = 1.455e13 / 398600 = 3.65e7 s^2 → √ = 6042 s
TOF = π × 6042 = 18,980 s ≈ 5.3 h (1st try: 10.5 h — used T not T/2)

RIGHT PAGE · CORNELL

CUES = questions **NOTES = keywords**

Why half an ellipse, not a straight line?

Which direction is each burn?

Fastest where? Which Kepler law?

Why r from Earth's CENTER?

? is Hohmann always cheapest?

orbit = falling + missing
Hohmann = 2 burns, both PROGRADE (outward)
burn1 @ perigee → apogee up to r2
burn2 @ apogee → perigee up, circular
K2: equal areas/time → fast @ perigee
K3: $T^2 \propto a^3$ · $T = 2\pi\sqrt{(a^3/\mu)}$ → eqn sheet
TOF = T/2 ← the classic mistake
r = altitude + 6371 !!
? C3 = launch energy → Q bank

SUMMARY — video closed, from memory

Hohmann = half an ellipse touching both circles. Two prograde burns: perigee stretches it out, apogee rounds it off. Takes half the ellipse's period: $\pi\sqrt{(a^3/\mu)}$, a=(r1+r2)/2, radii from Earth's center. Fastest at perigee (Kepler 2). LEO→GEO ≈ 5.3 h.

STEP 2 · TEMPLATE

Blank spread — copy these lines, then press play

Draw the dividers first. Dark boxes are what to do, not what to write.

🖉 LEFT PAGE · DRAW

Sec __ / Lec __ — title **date · pg __**

1 · Diagram coming? PAUSE. Sketch your guess in pencil. Big. Labeled.

2 · Instructor shows it? Redraw in pen next to your guess. Keep the wrong one.

3 · Bottom third: the equation worked by hand, IF there is one. No equation? A second diagram + the "3 takeaways" box go here instead. Key numbers → Equations & key numbers sheet.

❓ RIGHT PAGE · CORNELL

CUES (1/3) **NOTES (2/3)**

Every note → a QUESTION you'll answer from memory later.

**"What fixes GEO's altitude?"
not "GEO."**

Lost? Write "?" here. It goes to the Q bank and to Claude.

**Keywords · arrows · abbreviations.
NO full sentences — sentences = transcribing, not thinking.**

Box every equation.

Leave white space for review corrections.

🧠 SUMMARY — 5 lines, VIDEO CLOSED

Written last, from memory. Can't fill 5 lines? You don't have it yet. Good — that's the signal. Rewatch that bit, try again.

STYLE · PENCIL + HIGHLIGHTERS

Make it look good and still work

The popular "aesthetic notes" look is fine. The trap is decorating instead of thinking. Two rules fix that: every color has one fixed job, and color goes on AFTER, not during.

🖍 Highlighter key — pick once, never change

- **YELLOW** = a term/definition I must know
- **PINK** = an equation (box it in pencil, then pink)
- **BLUE** = a question / "?" / something to ask Claude
- **GREEN** = the "aha" — the one line that made it click
- **ORANGE** = a mistake I made → copy to Q bank

Only have 3? Keep yellow, pink, blue. Tape this key inside the front cover.

✏️ Pencil rules

- **Light pencil** = your guess (sketches, first-try math). **Dark pencil** = corrected. Don't erase the guess — that contrast is the memory.
- Highlighters are a **second pass**. Write the whole lecture in pencil first.
- **Highlight on the +1 day review**, not while writing. Color only what you could NOT recall when you covered the notes. Then the highlights literally mark your gaps — that's what makes them functional instead of decorative.

✅ Aesthetic things that help

- **Title banner** per lecture (boxed, shadow) — makes the page findable when flipping.
- **A tiny doodle icon** per topic (🚀 propulsion, 🛰 ADCS, 📡 comms) — a memory hook, 10 seconds max.
- **"3 takeaways" mini box** at the bottom of the DRAW page.
- Boxed equations with a worked number beside them.

🚫 Aesthetic things that steal study time

- Lettering practice, multiple fonts, re-copying "neat" versions, decorating during the video, highlighting whole paragraphs.

🖍 WHAT A FINISHED PAGE LOOKS LIKE (after the +1 day pass)

🪐 SEC 3 · LEC 2 — HOHMANN TRANSFER

11/9 · pg 13

Hohmann transfer = half an ellipse that touches both circular orbits
2 burns, both prograde (same direction you're moving)
burn 1 @ perigee → apogee up to r2
burn 2 @ apogee → perigee up → circular

TOF = $\pi \sqrt{(a^3/\mu)}$ a = (r1+r2)/2 μ = 398,600 km³/s²
TOF = T/2, NOT T — I did this wrong first try → Q bank

orbit = falling and missing. faster sideways = higher.
Kepler 2: equal areas in equal time → fastest at perigee
? is Hohmann always the cheapest transfer? → ask Claude
? what is C3?

⭐ **3 TAKEAWAYS**
1. Hohmann = 2 prograde burns, half an ellipse
2. TOF = half the period of the transfer ellipse
3. r is from Earth's CENTER (add 6371 km)

Notice: only ~6 things are highlighted on the whole page, each in the color that says what it is. You can read the page's structure from across the room, and the orange/blue marks are a to-do list.

STEP 3 · THE 70%

Come back. That's where it sticks.

Notes are ~30% of retention. Returning on a schedule is the other ~70%. Tape this inside the back cover.

🎬 During a lecture

- [ ] 1 Draw your guess **before** the diagram appears. Correct it after.
- [ ] 2 Equation or a number on screen? **Pause.** Cover it. Re-plug it yourself. (Few lectures will have one. That's fine.)
- [ ] 3 Keywords only in Notes. Questions only in Cues.
- [ ] 4 **Close the video.** Write the 5-line summary from memory.
- [ ] 5 Index line + any formula or key number to pages 3–4.

🏁 After a section

- [ ] 6 Do the quiz **cold** (Udemy's, or Claude's).
- [ ] 7 Every miss + every "?" → Question bank.
- [ ] 8 Tick "Watched" on the tracker.

⚡ Low-energy day? Minimum that still counts: the 5-line summary + one tick on the tracker. That's it. Done is done.

🔁 Spaced review · 5 min per section · tick the tracker

When	What
+1 day	Hand over the Notes column. Answer every Cue out loud.
+3 days	Same. Missed twice → Question bank.
+7 days	Same + 3 Q-bank questions from older sections.
+14 days	Same.
+30 days	Same. Clean at +30 = it's yours.

🗺 Weekly, 10 min

Blank page. Redraw the whole course map from memory:
mission → environment → orbit → propulsion → comms → ADCS → power.
One line of "what it is" per box. Compare to the index. Gaps → Q bank.

🤖 Using Claude while you study

- **Stuck mid-lecture:** pause, type the "?" in. Get the plain-English version + a diagram. Continue.
- **After a section:** photo of the spread, or type Cues + Summary. Claude checks for wrong facts and writes harder follow-ups for your Q bank.
- **Transcript button** (bottom-right of the Udemy player) → copy → paste for a gap check against your summary.